

TEJAS KUMAR S

AI/ML Engineer | Computer Vision • Generative AI & RAG • Deep Learning
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SUMMARY

AI/ML Engineer with hands-on experience in computer vision and generative AI, currently strengthening this foundation as an AI Engineer Intern at Datamites. Built and deployed transfer-learning image classifiers and end-to-end RAG pipelines using LangChain, Pinecone, and FastAPI. Backed by a strong analytical foundation from GATE preparation (2023–2025).

TECHNICAL SKILLS

Generative AI: LangChain, LangGraph, Prompt Engineering, Embeddings, Model Context Protocol (MCP), Agentic AI
RAG & Vector Search: Retrieval-Augmented Generation (RAG), Pinecone, FAISS, Vector Databases, Semantic Search
Computer Vision & Deep Learning: YOLO, VGG16, ResNet50, EfficientNet-B0, MobileNetV2, Transfer Learning & Fine-Tuning, CNN, ANN, OpenCV
APIs: FastAPI, RESTful APIs, Microservices Architecture
Deployment & DevOps: Docker, Docker Compose, Linux/WSL2, Prometheus, Monitoring & Observability
Machine Learning: Linear/Logistic Regression, Decision Trees, Random Forest, SVM, KNN, Naive Bayes, XGBoost
Core Programming & Frameworks: Python, SQL, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Git, GitHub, Streamlit

PROJECTS

Project 1: Intelligent RAG-Based Chatbot | [GitHub](#) | [Live Link](#)

Technology Used: Python, LangChain, Groq API, Pinecone, Streamlit, BeautifulSoup

- Problem:** Users had to manually dig through long PDFs, spreadsheets, and web pages to find specific information, which was slow and easy to get wrong.
- Approach:** Designed an end-to-end RAG pipeline that ingests PDFs, CSVs, and live web content, chunks and embeds it with Pinecone, and retrieves grounded context for a Groq-hosted LLM via LangChain.
- Result:** Shipped a streaming Streamlit chat interface with session analytics that gives fast, source-grounded answers and reduces hallucinated responses.

Project 2: Chest X-Ray Pneumonia Classification | [GitHub](#) | [Live Link](#)

Technology Used: Python, TensorFlow/Keras, OpenCV, Scikit-learn

- Problem:** Manual chest X-ray screening for pneumonia is slow and inconsistent, especially where radiologist availability is limited.
- Approach:** Applied transfer learning across MobileNetV2, EfficientNet-B0, and ResNet50, then diagnosed and fixed a majority-class collapse by optimizing for precision, recall, and false-negative rate instead of raw accuracy, since missed diagnoses are costliest.
- Result:** Deployed the best-performing model as a live, publicly accessible tool that gives a fast, consistent preliminary read to support clinical review.

Project 3: AI-Powered Incident Response Platform

Technology Used: Python, FastAPI, Docker, Docker Compose

- Problem:** Engineering teams often have no safe environment to rehearse coordinated incident response before something actually breaks in production.
- Approach:** Built a containerized microservices platform with FastAPI services communicating over REST APIs, orchestrated with Docker Compose on Linux/WSL2, with structured logging and HTTP timeout/error handling.
- Result:** Enables teams to run realistic incident simulations that validate inter-service communication and sharpen troubleshooting response time.

Project 4: Flight Fare Prediction

Technology Used: Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit

- Problem:** Travelers often overpay or misjudge flight costs because airline pricing fluctuates unpredictably.
- Approach:** Built an ML pipeline comparing Linear Regression, Decision Tree, Random Forest, and XGBoost, selecting the best model using R^2 , MAE, and RMSE with cross-validation.
- Result:** Delivered fare predictions through an interactive Streamlit app so travellers can estimate costs and plan budgets before booking.

EXPERIENCE

AI Engineer Intern | Datamites

Nov 2025 – Present

- Designed and deployed a RAG-based document search and Q&A system using LangChain, Pinecone, and FAISS to help retrieve accurate, context-grounded answers from large document sets instead of manual searching.
- Followed structured data preprocessing, feature engineering, and model evaluation practices across ML, DL, NLP, CV, and GenAI projects, maintaining clean version-controlled code with Git/GitHub for reproducible development.

EDUCATION

Bachelor of Engineering, Computer Science

Sri Krishna Institute of Technology, Bangalore | CGPA: 8.3

2019 – 2023

Pre-University Course

Ramaiah Composite Pre-University College, Bangalore | 73%

2016 – 2018

SSLC (CBSE)

Shree Ram Public School, Bangalore | CGPA: 8.8

2010 – 2015